HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and debugged **Linux Kernel device drivers** for embedded systems, focusing on memory management and inter-processor communication, directly contributing to SoC feature prototyping.
- Designed and implemented **ARM64 device drivers** for high-speed peripherals, ensuring efficient data transfer and system stability across mobile and compute chipsets.
- Collaborated with cross-functional teams including architects and verification engineers to integrate new SoC features, participating in sprint planning and technical estimation sessions.
- Wrote and optimized **C code for Linux Kernel modules**, focusing on data structures and algorithmic efficiency to enhance driver performance.
- Applied **ARM CoreSight debugging tools** to analyze system-level issues and optimize driver behavior in complex embedded environments.
- Participated in code reviews for embedded C/C++ driver development, providing feedback to peers on design patterns and kernel best practices.
- Containerized embedded development environments using **Docker**, enabling consistent build and test pipelines across different developer workstations.

SKILLS

Embedded Systems – Linux Kernel, Device Drivers, Embedded Systems Architecture, ARM CoreSight, ARM Architecture, RISC-V Architecture, DSPs, CPUs, High and Low Speed Peripherals, DDR, Interconnect, System Cache, QOS, Power Management, Boot, Debug, Security, Access Control Architecture

Programming Languages – C, C++, Python, Perl, Assembly

Operating Systems – Linux

Tools & Methodologies – SoC Development, Architecture, Data Structures, Git, CI/CD, Agile Methodologies, Debugging, Problem-Solving, Experimentation Methodologies

Cloud & Containerization – Docker

PROJECTS

Linux Kernel Driver for High-Speed Peripheral Interface

- Developed a **Linux kernel driver** in C for a custom high-speed peripheral interface, enabling data streaming at rates exceeding **5 GB/s**.
- Implemented advanced **data structures** within the driver to manage I/O buffers and minimize latency, reducing data loss by **~15%** in stress tests.
- Integrated the driver with the main kernel framework, ensuring compatibility with existing system calls and memory management subsystems.

SoC Feature Prototyping with ARM CoreSight

- Contributed to the prototyping of next-generation **System-on-chip (SoC) features** by developing and testing associated embedded drivers.
- Utilized **ARM CoreSight** for detailed debugging and performance analysis of driver interactions with the SoC architecture, identifying bottlenecks that improved execution efficiency by **~10%**.
- Documented driver design and testing procedures, creating reusable artifacts for future development cycles within the Core Platform team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)